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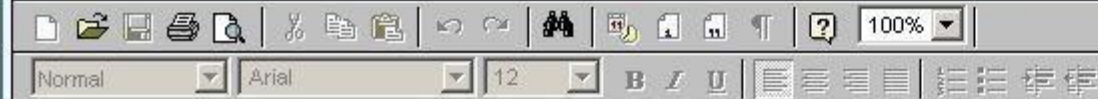
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Define cycle of a vibrating body?

Answer ([Please click here to Add Answer](#))



What is Logistic equation, write it mathematically?

Answer ([Please click here to Add Answer](#))



Write the solution $x(t) = c_1 \cos \omega t + c_2 \sin \omega t$ of a simple harmonic motion in a simple alternative form?

Answer ([Please click here to Add Answer](#))



Find the general solution of the following non-homogenous differential equation using annihilator operator

$$y'' - 3y' + 2y = 4x^2 + 5?$$

Answer ([Please click here to Add Answer](#))



Given that $y = c_1 e^x + c_2 e^{-x}$ is a two parameter family of solutions of the differential equation $\frac{d^2 y}{dx^2} - y = 0$ on $(-\infty, \infty)$ find a member of the family satisfying the boundary conditions $y(0) = 0, y'(1) = 1$

Answer ([Please click here to Add Answer](#))





In which sciences or subjects Logistic equations are useful?

Answer ([Please click here](#) to Add Answer)





Define cycle of a vibrating body?

Answer ([Please click here](#) to Add Answer)





What is Logistic equation, write it mathematically?

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Answer ([Please click here](#) to Add Answer)



Question No : 21 of 26

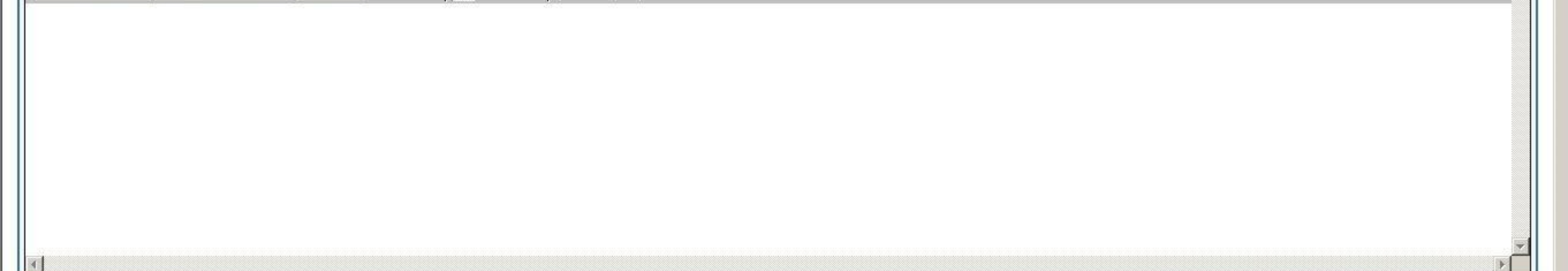
Marks: 2 (Budgeted Time 4 Min)

Find the order of the homogenous equation obtained from the following non-homogenous differential equation

$$y'' + 4y' + 3y = 4x^2 + 5?$$

Answer ([Please click here to Add Answer](#))

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21



Question No : 22 of 26

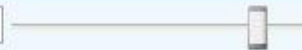
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If half-time life (T) of a radioactive isotope is 5 days, then find k.

Answer ([Please click here to Add Answer](#)) 

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Question No : 23 of 26

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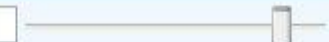
Determine the annihilator operator of the general solution of the following homogeneous linear first order differential equation:

$$\frac{dy}{dx} - 5y = 0$$

Answer ([Please click here to Add Answer](#)) 

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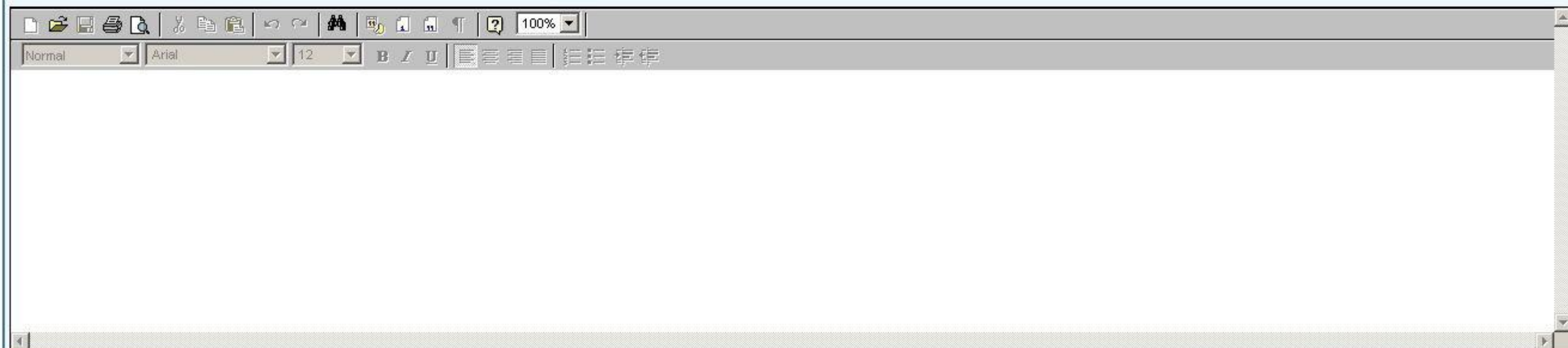
23 

Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min) 

If $y_1(x) = e^x$ is the first solution of $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 3y = 0$, then construct its second solution using $y_2(x) = y_1(x) \int \frac{e^{-\int P(x) dx}}{y_1^2(x)} dx$.

Note: where $P(x)$ is a coefficient of the first derivative involving in the above given differential equation.

Answer ([Please click here to Add Answer](#)) 

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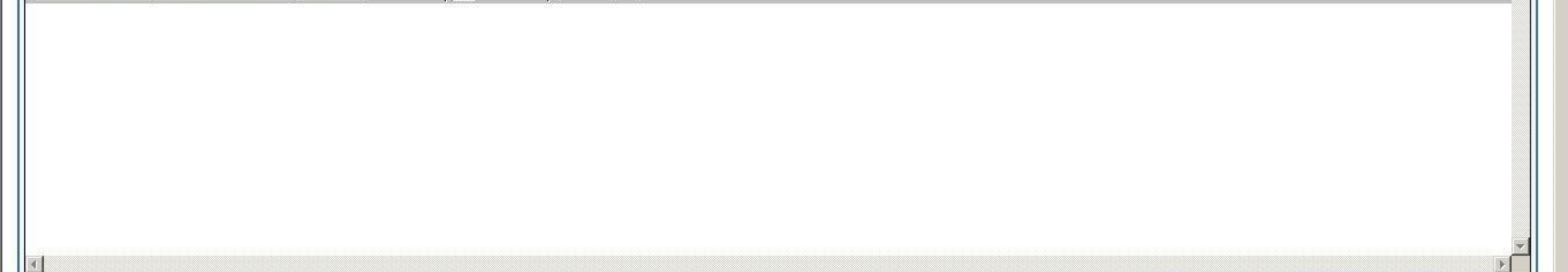
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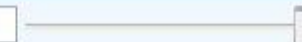
Find period of vibration of the solution of a simple harmonic motion?

Answer ([Please click here to Add Answer](#)) 

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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

Using Newton's Law of Cooling with initial condition $T(0)=T_0$ and if temperatures at time t_1 and t_2 are known then show that

$$\frac{T(t_1) - T_m}{T(t_2) - T_m} = e^{k(t_1 - t_2)}$$

Answer (Please [click here](#) to Add Answer)

If initial temperature of the cooling body is T_0 then we obtain the initial value problem

$$\frac{dT}{dt} = k(T - T_m), \quad T(0) = T_0$$

where k is constant of proportionality. The differential equation in the problem is linear as well as separable.

Separating the variables and integrating we obtain

$$\int \frac{dT}{T - T_m} = \int k dt$$

This means that

$$\ln |T - T_m| = kt + C$$

$$T - T_m = e^{kt+C}$$

$$T(t) = T_m + C_1 e^{kt} \quad \text{where} \quad C_1 = e^C$$

Now applying the initial condition $T(0) = T_0$, we see that $C_1 = T_0 - T_m$. Thus the solution of the initial value problem is given by

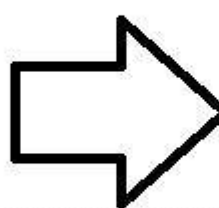
$$T(t) = T_m + (T_0 - T_m)e^{kt}$$

Hence, If temperatures at times t_1 and t_2 are known then we have

$$T(t_1) - T_m = (T_0 - T_m)e^{kt_1}, \quad T(t_2) - T_m = (T_0 - T_m)e^{kt_2}$$

So that we can write

$$\frac{T(t_1) - T_m}{T(t_2) - T_m} = e^{k(t_1 - t_2)}$$



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Evaluate $e^{-\int p(x) dx}$ using the differential equation $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + \left(x^2 - \frac{1}{4}\right)y = 0$ by writing it in the standard form $\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0$.

Answer ([Please click here to Add Answer](#))



If the complementary solution of the following differential equation is $c_1 \sin 2x + c_2 \cos 2x$, then what will be the general form of its particular solution:

$$\frac{d^2 y}{dx^2} + 4y = \cos 2x$$

Answer ([Please click here to Add Answer](#))



If a mass weighing $8lb$ and $k = 40 lb/ft$ then find the amount of elongation's'.

Answer ([Please click here to Add Answer](#))



If $y_1(x) = e^{-4x}$ is the first solution of $\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 8y = 0$, then construct its second solution using $y_2(x) = y_1(x) \int \frac{e^{-\int P(x) dx}}{y_1^2(x)} dx$.

Note: where $P(x)$ is a coefficient of the first derivative involving in the above given differential equation.

Answer ([Please click here to Add Answer](#))



A mass weighing $2lb$ stretches a spring 6 inches. Determine the equation of simple harmonic motion.

Answer ([Please click here to Add Answer](#))



A radioactive isotope has a half-life of 16 days. We have 45 g at the end of 32 days. How much radioisotope was initially present?

Answer ([Please click here to Add Answer](#))



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Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

If a mass weighing $20lb$ stretches a spring by $\frac{1}{2}ft$ then find k using the Hooke's Law.

Answer (Please [click here](#) to Add Answer)

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If a mass weighing $20lb$ stretches a spring by $\frac{1}{2}ft$ then find k using the Hooke's Law.

Answer ([Please click here to Add Answer](#))



Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min) 

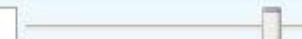
Show that $y = x^2$ is a solution of the following differential equation:

$$x^2 \frac{d^2 y}{d x^2} - 3x \frac{d y}{d x} + 4y = 0$$

Answer (Please [click here](#) to Add Answer) 

Rich text editor toolbar with icons for Bold, Italic, Underline, Text Color, Background Color, Bulleted List, Numbered List, Indent, Outdent, Link, Unlink, and others. The font is set to Arial, size 12, and the zoom level is 100%.

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Show that $y = x^2$ is a solution of the following differential equation:

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 0$$

Answer ([Please click here to Add Answer](#))



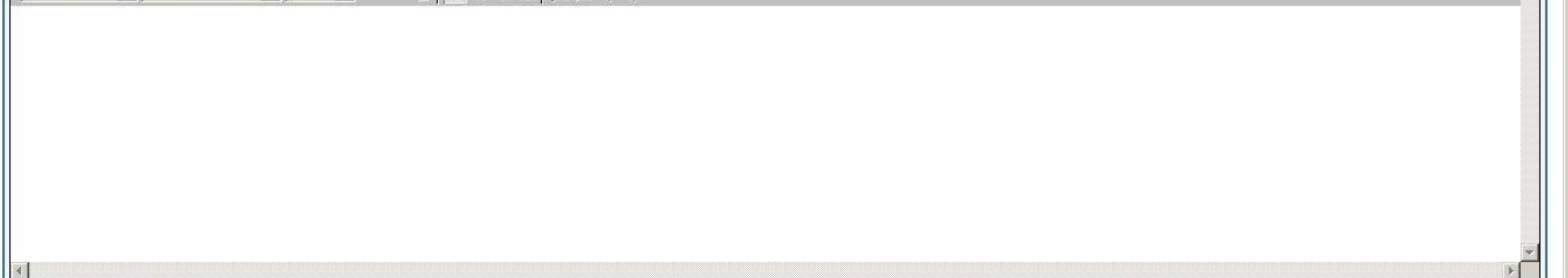
Question No : 23 of 26

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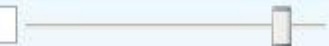
What is Logistic equation, write it mathematically?

Answer ([Please click here to Add Answer](#)) 

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23 

What is Logistic equation, write it mathematically?

Answer ([Please click here to Add Answer](#))



Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Write the following differential equation in the form $L(y) = g(x)$, where L is a linear differential operator with constant coefficients.

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 5y = 2 \sin y$$

Answer ([Please click here to Add Answer](#))

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Write the following differential equation in the form $L(y) = g(x)$, where L is a linear differential operator with constant coefficients.

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Answer ([Please click here to Add Answer](#))



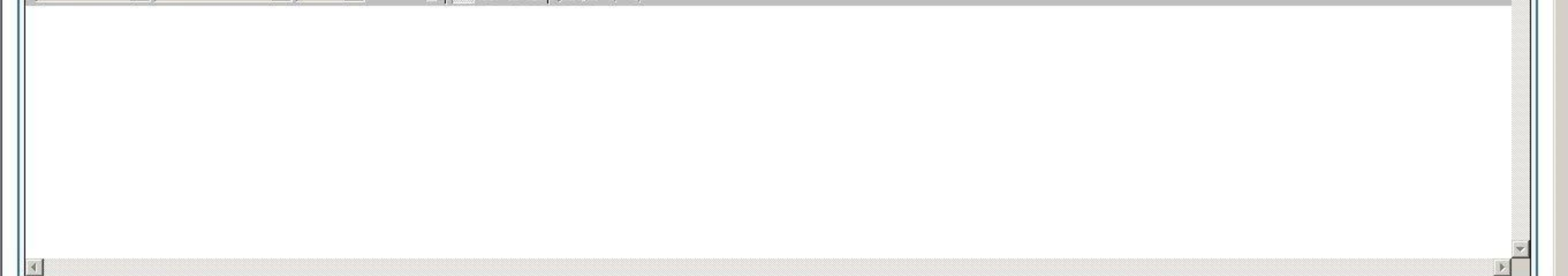
Question No : 25 of 26

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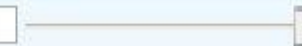
Find the wronskian of the Differential equation $y''' - 2y'' - 21y' - 18y = 3 + 4e^{-x}$ using variation of parameter and the root of the auxiliary equation is $m_1 = -3, m_2 = -1, m_3 = 6$?

Answer (Please [click here](#) to Add Answer) 

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Find the wronskian of the Differential equation $y''' - 2y'' - 21y' - 18y = 3 + 4e^{-t}$ using variation of parameter and the root of the auxiliary equation is $m_1 = -3, m_2 = -1, m_3 = 6$?

Answer ([Please click here to Add Answer](#))



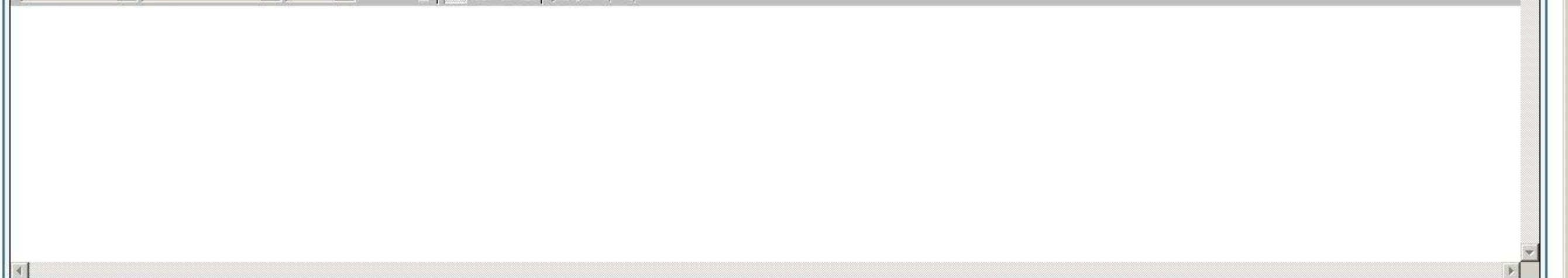
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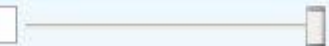
Solve the boundary value problem $\frac{d^2y}{dx^2} + 25y = 0$ subject to the boundary conditions $y(0) = 0$, $y\left(\frac{\pi}{10}\right) = 0$

Answer (Please [click here](#) to Add Answer) 

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Solve the boundary value problem $\frac{d^2y}{dx^2} + 25y = 0$ subject to the boundary conditions $y(0) = 0$, $y\left(\frac{\pi}{10}\right) = 0$

Answer ([Please click here to Add Answer](#))



Find the order of the homogenous equation obtained from the following non-homogenous differential equation

$$y'' + 4y' + 3y = 4x^2 + 5?$$

Answer ([Please click here to Add Answer](#))



If half-time life (T) of a radioactive isotope is 5 days, then find k.

Answer ([Please click here to Add Answer](#))



Determine the annihilator operator of the general solution of the following homogeneous linear first order differential equation:

$$\frac{dy}{dx} - 5y = 0$$

Answer ([Please click here to Add Answer](#))



If $y_1(x) = e^x$ is the first solution of $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 3y = 0$, then construct its second solution using $y_2(x) = y_1(x) \int \frac{e^{-\int P(x) dx}}{y_1^2(x)} dx$.

Note: where $P(x)$ is a coefficient of the first derivative involving in the above given differential equation.

Answer ([Please click here to Add Answer](#))



Find period of vibration of the solution of a simple harmonic motion?

Answer ([Please click here to Add Answer](#))



Using Newton's Law of Cooling with initial condition $T(0)=T_0$ and if temperatures at time t_1 and t_2 are known then show that

$$\frac{T(t_1) - T_m}{T(t_2) - T_m} = e^{k(t_1 - t_2)}$$

Answer ([Please click here to Add Answer](#))



Show that the functions $f_1(x) = \sin x$ and $f_2(x) = x$ are linearly independent.

Answer ([Please click here to Add Answer](#))



If the complementary solution of the following differential equation is $c_1 \sin 2x + c_2 \cos 2x$, then what will be the general form of its particular solution:

$$\frac{d^2 y}{dx^2} + 4y = \cos 2x$$

Answer ([Please click here to Add Answer](#))



Calculate the Wronskian of the following functions. Further determine whether they are linearly independent or not on the interval $(-\infty, \infty)$.

$$f_1(x) = \sin^2 x ; f_2(x) = \cos^2 x$$

Answer ([Please click here to Add Answer](#))



What will be the general form of the particular solution of the following differential equation:

$$\frac{d^2 y}{d x^2} - 2 \frac{d y}{d x} + y = e^x$$

Answer ([Please click here to Add Answer](#))



Deduce the Special Case of Logistic Equation "Epidemic spread"?

Answer ([Please click here to Add Answer](#))



Find the general solution of the following non-homogenous differential equation using annihilator operator

$$y'' - 3y' + 2y = 4x^2 + 5?$$

Answer ([Please click here to Add Answer](#))



Question No : 21 of 26

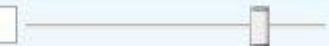
Marks: 2 (Budgeted Time 4 Min) 

If a mass weighing $20lb$ stretches a spring by $\frac{1}{2}ft$ then find k using the Hooke's Law.

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Question No : 22 of 26

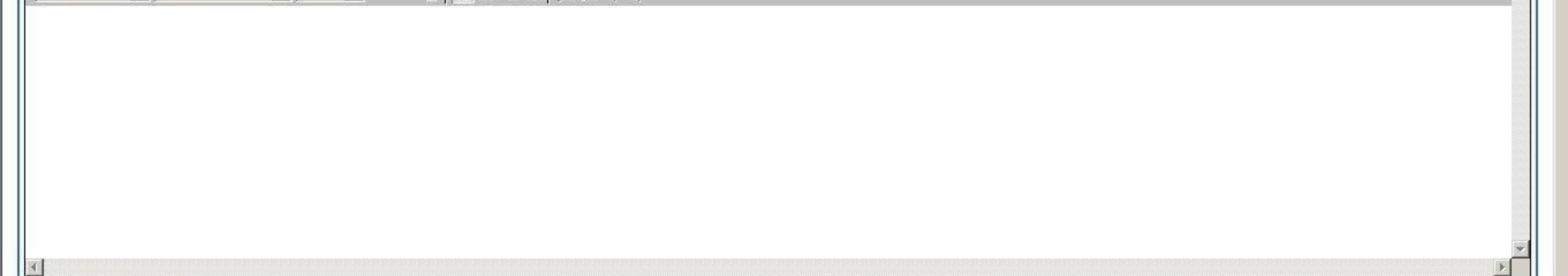
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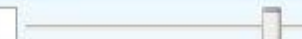
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Answer (Please [click here](#) to Add Answer) 

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Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min) 

What is Logistic equation, write it mathematically?

Answer ([Please click here to Add Answer](#)) 

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Question No : 24 of 26

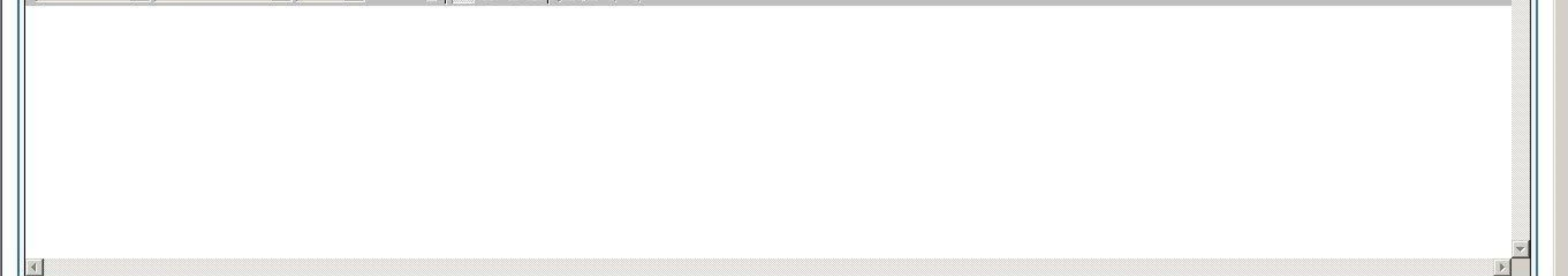
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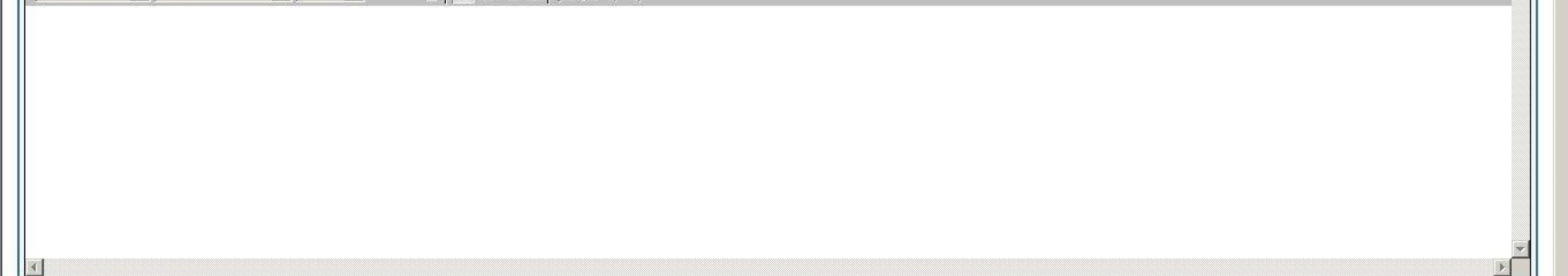
Question No : 25 of 26

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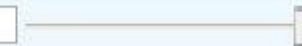
Find the wronskian of the Differential equation $y''' - 2y'' - 21y' - 18y = 3 + 4e^{-x}$ using variation of parameter and the root of the auxiliary equation is $m_1 = -3, m_2 = -1, m_3 = 6$?

Answer (Please [click here](#) to Add Answer) 

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Question No : 26 of 26

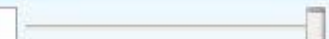
Marks: 5 (Budgeted Time 10 Min) 

Solve the boundary value problem $\frac{d^2y}{dx^2} + 25y = 0$ subject to the boundary conditions $y(0) = 0$, $y\left(\frac{\pi}{10}\right) = 0$

Answer (Please [click here](#) to Add Answer) 

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26 

Differential Equations (MTH401)

Question: **21** (Marks: 2)Attempted Questions: **25** Total Questions: **26**

Determine whether the following function is homogeneous or not? If not then justify the answer with appropriate reason:

$$f(x, y) = \frac{1}{y}$$

Differential Equations (MTH401)

Question: **22** (Marks: 2)Attempted Questions: **25** Total Questions: **26**

Find the characteristic (auxiliary) equation and its roots for the following homogeneous linear differential equation,

$$\frac{d^2 y}{dx^2} - 9y = 0$$

Answer:

Differential Equations (MTH401)

Question: **23** (Marks: 3)Attempted Questions: **25** Total Questions: **26**

A mass weighing 2 lb stretches a spring 6 inches. Determine the equation of the simple harmonic motion.

.....

Differential Equations (MTH401)

Question: **24** (Marks: 3)Attempted Questions: **25** Total Questions: **26**

Find the differential equation of the orthogonal family of curves $y = cx^2$, where c is an arbitrary constant.

.....

Answer:

Differential Equations (MTH401)

Question: **25** (Marks: 5)Attempted Questions: **25** Total Questions: **26**

Discuss the exactness of the following differential equation and solve it

$$(\sin y - y \sin x) dx + (\cos x + x \cos y) dy = 0$$

Answer:

Differential Equations (MTH401)

Question: **26** (Marks: 5)Attempted Questions: **25** Total Questions: **26**

Given $L = D^2 + 5D + 6$ is a linear differential operator. Then

1. Factorise L .

$$\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0$$

2. Verify that L annihilates the general solution of .

Answer: